

DO NOT WRITE ON TEST



**FINANCIAL MATH
& ANALYSIS CONCEPTS**

~OPEN EVENT~

190

NATIONAL 2025

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. For any interest calculations, assume a 360-day year.
5. Unless otherwise stated, assume that all plant assets were purchased on January 1.
6. Unless otherwise stated, assume that the overtime rate is 1.5 regular time pay for any hours worked in excess of 40 hours per week.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. A design firm bought a site license for creative software that retails for \$199.99 per copy. They paid \$159,990 for use on 900 computers. How much did they save by buying the site license?
 - A. \$20,001
 - B. \$30,001
 - C. \$40,001
 - D. \$50,001
2. A law firm bought a site license for legal research software that retails for \$250 per copy. They paid \$62,500 for the use of 300 lawyers and paralegals. How much did they save by buying the site license?
 - A. \$12,500
 - B. \$25,000
 - C. \$37,500
 - D. \$75,000
3. A marketing agency purchased a site license for analytics software that retails for \$99.99 per account. They paid \$79,992 for use on 1,000 accounts. How much did they save by buying the site license?
 - A. \$17,998
 - B. \$19,998
 - C. \$29,998
 - D. \$99,990
4. A multinational company acquired a site license for communication software that retails for \$9.99 per user. They paid \$79,920 for the use of 12,000 employees. How much did they save by buying the site license?
 - A. \$19,946
 - B. \$29,940
 - C. \$39,960
 - D. \$99,900
5. The Johnson family estimates their first year home operating expenses as follows: property tax, \$2,800; insurance, \$600; utilities, \$1,600; maintenance, \$1,000; mortgage interest, \$7,200; lost interest on down payment, \$500. Their estimated income tax savings are \$2,000. What will be the net cost of owning the home in the first year?
 - A. \$10,700
 - B. \$11,500
 - C. \$11,700
 - D. \$12,000
6. Sara earned \$7,500 last month. She paid an overall FICA tax rate of 7.65% that consists of a 6.2% social security tax and a 1.45% Medicare tax. Federal income taxes withheld from her wages were \$1,800. What total amount of taxes was deducted from her pay last month?

- A. \$573.75
 - B. \$2,373.75
 - C. \$1,800.00
 - D. \$3,373.75
7. Jake earned \$3,000 last month. He paid an overall FICA tax rate of 7.65% that consists of a 6.2% social security tax and a 1.45% Medicare tax. Federal income taxes withheld from his wages were \$612. What total amount of taxes was deducted from his pay last month?
- A. \$229.50
 - B. \$612.00
 - C. \$841.50
 - D. \$1,841.50
8. Lily earned \$8,000 last month. She paid an overall FICA tax rate of 7.65% that consists of a 6.2% social security tax and a 1.45% Medicare tax. Federal income taxes withheld from her wages were \$2,048. What total amount of taxes was deducted from her pay last month?
- A. \$612.00
 - B. \$2,048.00
 - C. \$2,660.00
 - D. \$4,660.00
9. The cost of a promotional flyer in a weekly magazine distributed to one area is \$2,000 per issue. There are 10,000 households in the area who receive the magazine. What will be the cost of the ad per household?
- A. \$0.18
 - B. \$0.20
 - C. \$0.22
 - D. \$0.25
10. John bought 300 shares of common stock at \$10.50 each. The commission on the purchase was \$125.50. John later sold the stock at \$15.25. The commission on the sale was \$150. What was the amount of profit or loss on the sale?
- A. \$1,100 profit
 - B. \$1,100 loss
 - C. \$1,149.50 profit
 - D. \$1,149.50 loss

11. Lisa purchased 200 shares of a company's stock at \$32.75. The commission on the purchase was \$98. Lisa later sold the stock for \$45.50. The commission on the sale was \$202. What was her profit or loss from the transaction?
- A. \$2,250 profit
 - B. \$2,250 loss
 - C. \$2,552 profit
 - D. \$2,552 loss
12. Noah is targeting a 20% annual net income on his \$8,000 investment in a small house. The annual expenses associated with the house are \$2,400. What monthly rent must Noah set to achieve his desired income?
- A. \$333.33
 - B. \$500.67
 - C. \$600.25
 - D. \$700.75
13. Ava seeks an 18% annual net income on her \$12,000 investment in a condominium. Her yearly expenses for the condo are \$3,000. What monthly rent is necessary for Ava to meet her financial goal?
- A. \$430
 - B. \$650
 - C. \$710
 - D. \$840
14. Denise is considering renting a flat for \$780 a month, with heat included. The security deposit will be one month's rent. She estimates additional annual costs to be: insurance \$120; utilities, \$540; parking, \$420. What is Denise's first-year cost of renting the flat?
- A. \$10,620
 - B. \$10,740
 - C. \$10,860
 - D. \$11,220
15. Ethan has decided to rent a loft for \$1,200 a month, including water and trash removal. The security deposit is one month's rent. His other estimated annual costs include insurance \$180; utilities, \$840; parking space, \$600. What is Ethan's first-year cost of renting the loft?
- A. \$15,020
 - B. \$15,820
 - C. \$16,220
 - D. \$17,220

16. Oliver had 30 square yards of laminate flooring installed in his office. The laminate cost \$8.99 per square yard, the underlayment cost \$3.50 per square yard, and installation was \$4.75 per square yard. The sales tax rate is 4.5%, applied only to materials. What was the total cost of the flooring?
- A. \$534.06
 - B. \$639.30
 - C. \$749.85
 - D. \$857.40
17. Chloe cashed in 250 shares of a mutual fund at \$18.80 per share. Her investment in the shares totaled \$4,000. What was her profit or loss?
- A. \$700 profit
 - B. \$700 loss
 - C. \$800 profit
 - D. \$800 loss
18. Ethan liquidated 100 shares of a mutual fund at \$25.30 per share. His investment in the shares was \$2,280. What was his profit or loss?
- A. \$220 profit
 - B. \$220 loss
 - C. \$250 profit
 - D. \$250 loss
19. Julia's checking account balance was \$2,050.60 at the start of the day. She made purchases of \$102.47, \$510.24, and \$365.89, and she deposited a check for \$750. What was Julia's balance at the end of the day?
- A. \$1,072.00
 - B. \$1,437.89
 - C. \$1,822.00
 - D. \$1,948.13
20. The Smiths' gross income for a year was \$85,000. From that amount, they subtracted a standard deduction of \$12,400, and 4 exemptions at \$4,200 each to find their taxable income. What was their taxable income total?
- A. \$52,000
 - B. \$55,800
 - C. \$60,000
 - D. \$64,400
21. Jackson's gross annual income was \$93,500. He subtracted a standard deduction of \$10,300, and 2 exemptions at \$4,150 each to calculate his taxable income. What was his taxable income?
- A. \$10,300
 - B. \$74,900
 - C. \$78,200
 - D. \$83,200

22. Chloe's gross income for a year was \$50,000. After subtracting a standard deduction of \$8,000 and 1 exemption at \$3,500, what was her taxable income?
 - A. \$38,000
 - B. \$38,500
 - C. \$42,000
 - D. \$43,500
23. The Lopez family's gross income for the year was \$150,000. From this, they subtracted a standard deduction of \$11,800 and 5 exemptions at \$3,800 each to find their taxable income. What was it?
 - A. \$119,200
 - B. \$121,000
 - C. \$128,200
 - D. \$130,000
24. Green Landscaping sold these items to a garden center: 30 bags of soil @ \$15.99 a bag, 20 gallons of fertilizer @ \$27.50 a gallon, and 15 hedge trimmers @ \$89.99 each. What was the total of the sales invoice before tax?
 - A. \$1,349.85
 - B. \$1,784.35
 - C. \$2,379.55
 - D. \$4,184.25
25. Fresh Bakery supplied these items to a café: 100 loaves of bread @ \$2.49 each, 50 cakes @ \$15.00 each, and 25 coffee bags @ \$9.99 each. What was the total of the sales invoice before tax?
 - A. \$1,623.50
 - B. \$1,248.75
 - C. \$1,873.25
 - D. \$2,498.75
26. A store purchased hats to sell for \$25 and sold them for \$35. If the markup is based on cost, what markup percentage is the store charging?
 - A. 20%
 - B. 30%
 - C. 40%
 - D. 50%
27. A boutique bought dresses to sell for \$150 and sold those dresses for \$225. If the markup is based on cost, what markup percentage is the boutique charging?
 - A. 25%
 - B. 40%
 - C. 50%
 - D. 60%

28. Tom is paid \$1.75 for every quality item he packs. For one week, he packed 300 items, 20 of which were rejected for poor quality. What was Tom's gross pay for the week?
- A. \$490.00
 - B. \$525.75
 - C. \$560.50
 - D. \$490.00
29. Carla is paid \$4.00 for every acceptable article she writes. Last week, she wrote 100 articles, 5 of which did not meet the publication standards. What were Carla's total earnings for the week?
- A. \$310.00
 - B. \$350.00
 - C. \$380.00
 - D. \$420.00
30. Miguel makes \$5.50 for each quality graphic he designs. During a productive week, he designed 200 graphics, but 15 were unusable. What was Miguel's gross pay for the week?
- A. \$1,050.25
 - B. \$1,100.00
 - C. \$1,017.50
 - D. \$1,185.00
31. Clara listed these items on her checking account deposit slip: (bills) 30 ones, 7 fives, 4 tens, and 8 twenties; (coins) 100 quarters, 50 dimes, 20 nickels; one check for \$55.75. What was the total of Clara's deposit?
- A. \$295.75
 - B. \$345.25
 - C. \$351.75
 - D. \$445.25
32. Lucas listed these items on his checking account deposit slip: (bills) 34 ones, 2 fives, 3 tens, and 4 twenties; (coins) 112 quarters, 67 dimes, 25 nickels; one check for \$38.25. He received cash of one ten-dollar bill. What was the total of Lucas's deposit?
- A. \$210.40
 - B. \$228.20
 - C. \$310.70
 - D. \$360.50
33. Hannah listed these items on her checking account deposit slip: (bills) 42 ones, 8 fives, 6 tens, and 9 twenties; (coins) 135 quarters, 80 dimes, 30 nickels; one check for \$67.00. What was the total of Hannah's deposit?
- A. \$425.50
 - B. \$432.25
 - C. \$525.50
 - D. \$575.00

34. You place \$750 in a three-year savings account that pays 4% compounded monthly. What is the effective annual interest rate, to the nearest tenth percent?
- A. 3.90%
 - B. 4.05%
 - C. 4.10%
 - D. 5.00%
35. You invest \$2,000 in a one-year bond that pays 7% compounded annually. What is the effective annual interest rate, to the nearest tenth percent?
- A. 6.80%
 - B. 7.00%
 - C. 7.20%
 - D. 7.50%
36. You save \$1,500 in a one-year fixed deposit that pays 8% compounded semi-annually. What is the effective annual interest rate, to the nearest tenth percent?
- A. 8.1%
 - B. 8.2%
 - C. 8.3%
 - D. 8.4%
37. Calculate the property tax due on a house assessed at \$250,000 if the tax rate is \$25.50 per \$1,000.
- A. \$5,625.00
 - B. \$5,875.00
 - C. \$6,125.00
 - D. \$6,375.00
38. Determine the property tax on a condo assessed at \$425,000 if the tax rate is \$40.75 per \$1,000.
- A. \$16,456.25
 - B. \$16,743.75
 - C. \$17,031.25
 - D. \$17,318.75
39. Liam bought a computer for \$1,200. He anticipates using it for a total of 5 years, with an estimated residual value of \$200. What depreciation should Liam record for the first year?
- A. \$150.00
 - B. \$190.00
 - C. \$200.00
 - D. \$320.00

40. Maya purchased a piece of machinery for \$50,000. She expects the machinery to produce 150,000 units over its lifetime and to have a salvage value of \$5,000. If the machinery produced 20,000 units in the first year, how much depreciation will Maya record for that year?
- A. \$4,000.00
 - B. \$5,000.00
 - C. \$6,000.00
 - D. \$7,000.00
41. Alex bought 300 shares of stock at a total cost of \$9,450. He sold the shares at \$35.50 and was charged a commission and other fees of \$300. What was his profit or loss from the sale of the stock?
- A. \$700 profit
 - B. \$700 loss
 - C. \$900 profit
 - D. \$900 loss
42. Ethan bought 250 shares of stock at a total cost of \$7,375. He sold the shares at \$32.40 and was charged a commission and other fees of \$250. What was his profit or loss from the sale of the stock?
- A. \$375 profit
 - B. \$375 loss
 - C. \$475 profit
 - D. \$475 loss
43. Mark's pension plan pays him (2% times year of service) times his average salary for the last three years. In the last three years, he earned \$60,000, \$62,500, and \$65,000. He will be retiring after 25 years of service. What monthly pension will he receive?
- A. \$2,604.17
 - B. \$2,708.33
 - C. \$2,875.00
 - D. \$3,041.67
44. Susan's retirement plan offers her 2.5% of her average salary over the last three years for each year of service. Her earnings for those years were \$55,000, \$58,000, and \$60,500. After working for 35 years, what will her monthly pension be?
- A. \$3,046.88
 - B. \$3,183.33
 - C. \$3,319.79
 - D. \$4,217.01

45. Last year, Brazil had exports of \$45.23 billion and imports of \$27.91 billion. How much of a deficit or surplus did they have?
- A. \$17.32 billion deficit
 - B. \$17.32 billion surplus
 - C. \$18.32 billion deficit
 - D. \$18.32 billion surplus
46. This year, Canada had exports of \$525.4 billion and imports of \$498.7 billion. How much of a deficit or surplus did they have?
- A. \$25.7 billion deficit
 - B. \$25.7 billion surplus
 - C. \$26.7 billion deficit
 - D. \$26.7 billion surplus
47. Marcus attended a 3-day workshop. He traveled 220 miles to and from the workshop by personal car at \$0.42 a mile. He was also given a \$160 per diem per full day (Marcus does not receive a per diem for partial days). What was his total reimbursement for the workshop?
- A. \$425.60
 - B. \$512.40
 - C. \$572.40
 - D. \$632.40
48. An online retailer paid \$4,000 for an e-commerce site setup and \$75.75 a month for maintenance. The annual domain registration fee is \$30, and the monthly fee for server space and internet connection is \$80. What is the total cost for the first year?
- A. \$5,279
 - B. \$5,609
 - C. \$5,899
 - D. \$6,269
49. Sophie invested \$4,000 in a 3-year CD that paid 3.8% annual interest. When she withdrew \$1,000 at the end of 2 years, her early withdrawal penalty was 6 months' interest. What was the amount of the penalty?
- A. \$38.00
 - B. \$57.00
 - C. \$76.00
 - D. \$114.00
50. Dave works on a piece-rate basis. He completed 50 pieces on Monday, 55 on Tuesday, 60 on Wednesday, and 65 on Thursday. He is paid \$1.25 for each piece. How many pieces must he complete on Friday so that his earnings for the 5 days will average \$70 a day?
- A. 50
 - B. 80
 - C. 85
 - D. 90